

Industry Solutions Framework

Cortex Code Skills Development Lifecycle

Health Sciences
Industry Solutions Tiger Team

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Executive Summary

This document defines the Industry Solutions Framework for developing, hardening, and distributing Cortex Code skills tailored to Health Sciences (covering both healthcare and life sciences). The framework establishes a two-repo model separating rapid experimentation from production-grade skill delivery, with a Tiger Team serving as the quality gateway.

The end goal: Field teams run a single command to get a curated, tested set of health sciences skills that solve standard problems across the industry.

```
cortex profile add health-sciences-solutions
```

This single command delivers all approved health sciences skills, an orchestrator that routes across 7 business domains, Cortex Knowledge Extensions for PubMed and ClinicalTrials.gov, and data model knowledge repositories - all validated through a rigorous promotion pipeline.

Two Profiles

The framework uses two distinct profiles to support different stages of the lifecycle:

	Incubator Profile	Production Profile
Name	health-sciences-incubator	health-sciences-solutions
Purpose	Test experimental skills during Phase 1	Deliver approved production skills
Skills source	Incubator repo (all draft skills)	SFS skills repo (production only)
Who uses	Contributors & testers	Field teams for customer engagements
Quality	Experimental - may be unstable	Validated, tested, production-grade
Command	cortex profile add health-sciences-incubator	cortex profile add health-sciences-solutions

Two-Repo Model

The framework uses two distinct repositories to separate experimentation from production:

	Incubator Repo	SFS Skills Repo
Repo	Snowflake-Solutions/health-sciences-incubator	Snowflake-Solutions/cortex-code-skills
Purpose	Rapid skill development & experimentation	Approved, production-grade skills
Who contributes	Anyone (SEs, SAs, field)	Tiger Team only
Branch/PR rules	Open branching, lightweight review	Formal PR, CI gates, promotion stages
Quality bar	Low - iterate fast, break things	High - best practices, evidence, testing
Skills flow	Skills created & refined here	Mature skills graduate here from incubator

Phase 0: Framework Setup

Accountable: Tiger Team

Tiger Team establishes the foundation for the entire lifecycle.

- Create industry skills INCUBATOR repo in Snowflake-Solutions GitHub
e.g., Snowflake-Solutions/health-sciences-incubator
Open access - any SE/SA can branch, commit, iterate
- Define skill development guidelines & templates for incubator
- Define quality bar & promotion criteria for graduation to SFS skills repo
- Set up lightweight CI in incubator (structure checks, linting - no blocking gates)
- Create incubator profile (health-sciences-incubator)
References incubator repo skills for easy testing
Publish to Snowflake registry for contributor access
- Document the full lifecycle: incubator -> SFS skills repo -> profile -> field

Output: Incubator repo, incubator profile, guidelines, promotion criteria, lifecycle docs

Phase 1: Incubate & Iterate

Accountable: Individual Contributors (SEs, SAs, Field Teams)

Repo: Snowflake-Solutions/health-sciences-incubator

Any SE, SA, or field team member can contribute to the incubator repo. This is a sandbox for rapid iteration with no formal approval gates.

- Branch freely, no formal PR approval needed
- Create, test, break, refine skills collaboratively
- Share work-in-progress with peers for feedback
- Use skill-development skill to scaffold new skills
- Test skills via incubator profile:
cortex profile add health-sciences-incubator
Profile auto-loads all incubator skills for testing
cortex profile sync health-sciences-incubator to get latest changes
- Or test individually: cortex skill add /path/to/incubator/skills/my-skill
- When skill is mature -> signal to Tiger Team for Phase 2 pickup

Gate: None - this is a sandbox for rapid iteration

Exit Criteria: Contributor believes skill is ready, notifies Tiger Team

Phase 2: Harden & Promote

Accountable: Tiger Team Only

Repo: Snowflake-Solutions/cortex-code-skills

Tiger Team picks up mature skills from the incubator and drives them through the full SFS approval pipeline. Only Tiger Team members can submit and merge promotion PRs in the SFS skills repo.

Step 1: Evaluate & Accept

- Review skill from incubator for relevance & quality potential
- Decide: accept, request changes, or decline
- Fork/copy skill into Snowflake-Solutions/cortex-code-skills

Step 2: Audit & Refine

- Audit against SKILL_BEST_PRACTICES.md
- Refine workflows, stopping points, guardrails
- Ensure skill composes well with other industry skills
- Validate no overlap/conflict with bundled product skills

Step 3: Test & Validate

- Test with realistic customer scenarios
- Gather testers (3+ for staging, 5+ for production)
- Record customer impact evidence (skill_evidence.yaml)
- Run overlap detection (blocking at staging+)

Step 4: Submit & Promote via PR

- Submit PR to Snowflake-Solutions/cortex-code-skills at 'draft' stage
- Pass full CI pipeline (evidence, overlap, best practices)
- Promote: draft -> review -> staging -> production
- Only Tiger Team members can merge promotion PRs

Step 5: Integrate into Production Profile

- Add approved skill to health-sciences-solutions profile
- Update orchestrator/router if needed
- Wire CKEs, data model knowledge, cross-domain patterns

Promotion Stages

Stage	Min Testers	Min Customers (cust-facing)	Overlap	Best Practices
draft	0	0	Warnings	Warnings
review	1	0	Warnings	Warnings
staging	3	1	Blocking	Warnings
production	5	1	Blocking	Blocking

Phase 3: Publish & Distribute

Accountable: Tiger Team

Tiger Team publishes the industry profile to the Snowflake registry, making it available to all field teams with a single command.

- Production profile (health-sciences-solutions) references only production-stage skills
- Includes orchestrator system prompt (domain routing, CKE integration)
- Publish to Snowflake registry: cortex profile publish health-sciences-solutions
- Maintain incubator profile separately for ongoing experimentation
- Document prerequisites (Marketplace CKEs, Cortex Search setup, etc.)
- Announce availability to field organization

Output: Published profile in Snowflake registry, onboarding docs

Phase 4: Consume & Feedback

Accountable: Field Teams (consumers), Tiger Team (support & updates)

Field SEs and SAs consume the published profile for customer engagements and feed improvements back into the incubator.

- cortex profile add health-sciences-solutions
- Uses approved, production-grade skills for customer engagements
- Files issues / feedback on existing skills
- Contributes improvements back to INCUBATOR -> feeds Phase 1
- cortex profile sync health-sciences-solutions for updates
- Can also add incubator profile to preview upcoming skills:
cortex profile add health-sciences-incubator

Feedback loop: Field issues -> incubator repo -> Tiger Team triage -> Phase 2

Skill Taxonomy

Skills are organized in a five-level hierarchy that maps from the broadest industry classification down to individual capabilities.

Hierarchy Definition

Industry / Sub-Industry / Business Function / Use Case Skill / Sub-Skill

Level	Example	Description	Applies To
Industry	Health Sciences	Top-level domain	Both repos
Sub-Industry	Provider, Pharma, Payer	Customer segment	Incubator structure
Business Function	Clinical Research, Drug Safety	Functional area within sub-industry	Incubator structure
Use Case Skill	healthcare-imaging, pharmacovigilance	Concrete skill solving a use case	Both repos
Sub-Skill	dicom-parser, imaging-viewer	Leaf capability within a skill	Skill internal

Sub-Industries

Sub-Industry	Customer Types	Business Functions
Provider	Hospitals, health systems, clinics, IDNs	Clinical Research, Clinical Data Management, Revenue Cycle
Pharma	Pharma, biotech, CROs	Drug Safety, Genomics, Lab Operations
Payer	Health plans, TPAs, PBMs	Claims Processing (future skills)
Cross-Industry	All of the above	Research Strategy, Knowledge Extensions

Incubator Repo Structure

The incubator repo uses the full hierarchy as directory nesting. This provides clear organization for browsing and discovery during development.

```
skills/
  health-sciences/
    provider/
      clinical-research/
        healthcare-imaging/
          dicom-parser/
          dicom-ingestion/
          dicom-analytics/
          imaging-viewer/
          imaging-governance/
          imaging-ml/
        clinical-data-management/
          fhir-data-transformation/
          clinical-nlp/
          omop-cdm-modeling/
        revenue-cycle/
          claims-data-analysis/
    pharma/
      drug-safety/
        pharmacovigilance/
        clinical-trial-protocol/
      genomics/
        nextflow-development/
        variant-annotation/
        single-cell-rna-qc/
        scvi-tools/
        survival-analysis/
      lab-operations/
        instrument-data-to-allotrope/
    payer/
      claims-processing/
  cross-industry/
    research-strategy/
      scientific-problem-selection/
    knowledge-extensions/
      cke-pubmed/
      cke-clinical-trials/
```

Leaf-level skills can appear directly under a business function (e.g., claims-data-analysis under revenue-cycle) without requiring an intermediate use-case-skill directory.

SFS Repo Naming Convention

The SFS skills repo (Snowflake-Solutions/cortex-code-skills) uses a flattened naming convention. Skills sit in a single skills/ directory with a structured prefix that encodes the taxonomy.

Naming Pattern

hcls-{sub-industry}-{function}-{skill}

Naming Rules

Component	Values	Notes
hcls	Fixed prefix	Health Sciences industry identifier
sub-industry	provider, pharma, payer, cross	Maps to sub-industry taxonomy level
function	imaging, cdata, dsafety, genomics, lab, claims, rwe, research, cke	Short alias for business function
skill	dicom-parser, pharmacovigilance, nextflow, etc.	Use case skill name (leaf level)

Example Mappings

Incubator Path	SFS Name
health-sciences/provider/clinical-research/healthcare-imaging/dicom-parser/	hcls-provider-imaging-dicom-parser
health-sciences/pharma/drug-safety/pharmacovigilance/	hcls-pharma-dsafety-pharmacovigilance
health-sciences/pharma/genomics/variant-annotation/	hcls-pharma-genomics-variant-annotation
health-sciences/provider/clinical-data-management/clinical-nlp/	hcls-provider-cdata-clinical-nlp
health-sciences/cross-industry/knowledge-extensions/cke-pubmed/	hcls-cross-cke-pubmed
health-sciences/provider/revenue-cycle/claims-data-analysis/	hcls-provider-claims-data-analysis

Orchestrator Routing Instructions

The health-sciences-solutions orchestrator profile uses the following rules to route requests to the correct skills:

- Route by sub-industry FIRST: Hospital/clinic -> Provider, Pharma/biotech/CRO -> Pharma, Health plan/TPA/PBM -> Payer

- When sub-industry is ambiguous: route by TASK, not customer type (e.g., imaging tasks always go to Provider > Clinical Research)
- Cross-industry skills (CKEs, research strategy) are available to ALL sub-industries
- Accept overlaps: some skills serve multiple sub-industries - route to the skill regardless of where it sits in the taxonomy
 - claims-data-analysis: serves both Provider and Payer
 - survival-analysis: serves both Pharma and Provider
 - clinical-nlp: serves both Provider and Pharma

Accountability Matrix

Phase	Who Can Do	Who Approves	Repo	Output
0 - Framework Setup	Tiger Team	Tiger Team	Both	Repos, guidelines, CI
1 - Incubate & Iterate	Anyone	Self-service	Incubator	Experimental skills
2 - Harden & Promote	Tiger Team only	Tiger Team	SFS Skills	Production-grade skills
3 - Publish & Distribute	Tiger Team	Tiger Team	N/A (registry)	Published profile
4 - Consume & Feedback	Field teams	Tiger Team (support)	Incubator (feedback)	Solutions, feedback

Lifecycle Flow

